

Gen AI Architect Program

Assignment — Student Grade System | learn. build. operate

1. What you will build

Write a Python program that asks the user to enter a mark (a number between 0 and 100) and then prints the matching letter grade. The program runs in the terminal — there is no website or app interface. You will set up a clean Python environment, write your code in a `.py` file, run it, and confirm it produces the correct output. Use pure Python only — no frameworks or external packages.

2. Grading scale to implement

Your program must convert a mark into a grade using exactly this scale:

Mark range	Grade
90 – 100	A
80 – 89	B
70 – 79	C
60 – 69	D
Below 60	E

Note: the boundaries are inclusive — a mark of exactly 90 is an A, exactly 80 is a B, and so on.

3. Requirements

- Use **only** the Python standard library. Do not install or import any third-party package or framework.
- Take the mark as input from the user while the program is running.
- Print a clear message showing the entered mark and the resulting grade.
- Handle the full 0–100 range and cover every grade band above.
- Keep all your code in a single file named `grade_system.py`.

Think about edge cases

What should happen if someone types a number above 100, below 0, or something that is not a number at all? Decide how your program should respond, and make sure it does not crash unexpectedly.

4. Step-by-step setup

Follow these steps in order. Commands are shown for both macOS/Linux and Windows where they differ.

Step 1 — Create a project folder

```
mkdir grade-system
cd grade-system
```

Step 2 — Create a virtual environment (venv)

A virtual environment keeps this project isolated from the rest of your system.

```
python -m venv venv
# (on some systems use 'python3' instead of 'python')
```

Step 3 — Activate the virtual environment

```
# macOS / Linux:
source venv/bin/activate

# Windows (PowerShell):
venv\Scripts\Activate.ps1

# Windows (Command Prompt):
venv\Scripts\activate.bat
```

When activated, you will see (venv) at the start of your terminal prompt.

Step 4 — Run your program

```
python file_name.py
```

Test it several times with different marks to confirm each grade band works.

Step 5 — Deactivate when finished

```
deactivate
```

5. Example of how a run should look

Your wording can differ, but a single run should behave roughly like this:

```
Enter your mark (0-100): 85
Mark: 85 -> Grade: B
```

6. What to submit

- Your `grade_system.py` file.
- A short screenshot or paste of the terminal showing the program running with at least three different marks.
- Two or three sentences explaining how you handled invalid input.
- Upload the code to GitHub and send us the repository link.

Reminder: this is a solo task using pure Python only. Build the logic yourself — the steps above are your roadmap, not the answer. Good luck!